Agilent B1500A
Semiconductor Device Analyzer

Ten Nanosecond Pulsed IV Parametric Test Solution

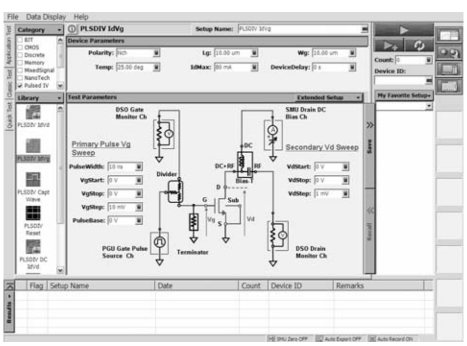
Proven, Accurate High-k/SOI characterization by 10nsec pulse width

Technical Overview

June 2006

Overview

State of the art semiconductor processes continue to meet market demands requiring smaller lithography, faster switching times, and lower power consumption. Accompanying the transition to the 45 nanometer node is the use of high-k gate dielectrics and silicon-on-insulator (SOI) transistors. The Agilent B1500A is well-positioned to act as the focal point for the parametric characterization of these devices in laboratory environments. Already possessing superb dc measurement performance, the B1500A with EasyEXPERT software now supports a pulsed IV measurement solution utilizing an Agilent 81110A pulse pattern generator and Agilent DSO80000B /54850 series scope that can characterize MOSFETs with an unprecedented gate pulse width of 10 nanoseconds.



Application Library on the B1500A EasyEXPERT

Features

- **10 nanosecond gate pulse widths**

This solution provides a 10 ns gate pulse width with 2 ns rise and fall times. It produces a clean rectangular waveform with minimal overshoot and undershoot.

- **1μA current measurement resolution**

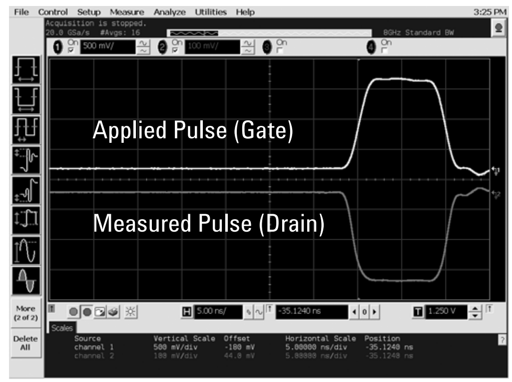
This solution gives 1μA current measurement resolution using a 10 ns pulse width, enabling more precise characterization of high-k and SOI devices.

- **Feedback loop enables accurate ID-VD and ID-VG measurement**

This solution monitors the actual MOSFET drain voltage and automatically corrects the applied drain voltage to insure that the actual drain voltage (on the transistor side of the load resistor) is correct for each measurement point. This is supported for both ID-VD and ID-VG measurements.

- **Correlation with production test environment**

This well-proven solution is available for and already being used on the 4075 and 4076 test systems. This makes it easy to correlate results measured in the lab with data taken on the production line.



Applied 10nsec gate pulse and measured drain current

- **Easy switching between dc and pulsed measurements**

A dc to pulsed IV switching option is available to enable you to toggle between dc measurements and pulsed IV measurements without having to change any cables. This makes it easy to correlate dc measurements with pulsed measurements, and it also permits the automation of this process.

- **Can reuse existing instruments to lower costs**

If you already have the Agilent pulse generator and oscilloscope you can use them with the pulsed IV solution. This reduces the initial cost of the solution.

- **Easy setup using Agilent EasyEXPERT software**

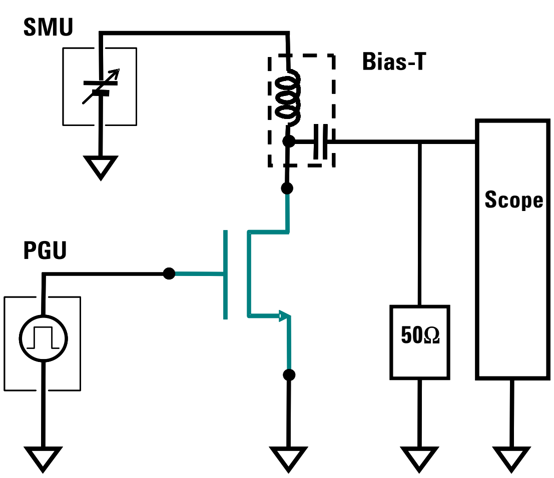
Agilent EasyEXPERT software, which is included on the B1500A and is also available in a desktop version, makes it easy for even neophyte users to make pulsed IV measurements. After selecting a pulsed IV application test, an intuitive GUI-based test setup window displays a complete schematic of the test equipment, making it easy to connect the components correctly. Following a straightforward "fill-in-the-blanks" process, the user clicks the "measure" button and begins making pulsed IV measurements. A graph and list of the data are generated automatically, and the user can export the data into a variety of data-analysis tools such as Microsoft Excel.

Typical Technical Information

Gate Pulse Width : 10 ns to 1 μs
Gate Pulse Voltage : -4.5V to 4.5V
Drain Pulse Measurement Current: Max. 80mA
Drain Pulse Measurement Current Resolution : 0.1μA
Drain Voltage Range : -10V to 10V

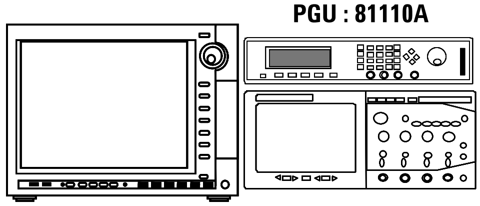
Block Diagram for Ultra-Short Pulsed IV

This block diagram shows a basic overview of the pulsed IV measurement. The drain voltage (or calculated current going into the drain) of the MOSFET is measured by the oscilloscope while an ultra-short pulse is applied to the gate by the PGU. The benefits of this for each technology type are:



1. For SOI transistors, the short duty cycle reduces self-heating effects that adversely impact the measurement results.
2. For high-k gate dielectric transistors, the short duty cycle reduces the incidence of electron trapping that distorts measurement results.

System Configuration



Operation Environment

Supported Instrument

Agilent B1500A
Semiconductor Device Analyzer

Agilent 81110A
Pulse Pattern Generator

Agilent 54850 Series / DSO80000B Series
Infinium Oscilloscope

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